



Republic of Malawi

Syllabus for

Agriculture

Forms 3 and 4

Ministry of Education, Science and Technology

Syllabus for

Agriculture

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Ministry of Education, Science and Technology

Prepared and published by

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Foreword

Education is the vehicle through which every citizen can realise his or her potential and contribute to national development. The vision of the education sector in Malawi is to be a catalyst for socio-economic development, industrial growth and an instrument for empowering the poor, the weak and the voiceless. Its mission is to provide quality and relevant education to Malawians. As a catalyst for the development, education should equip the individual with knowledge, skills, values and attitudes to enable one to perform one's roles effectively, in an attempt to promote and sustain the social-economic development of a nation.

It is the conviction of the Ministry of Education, Science and Technology that primary education alone is not sufficient for achieving socio-economic development. As argued in the NESP (2008-2017), 'In an evolving and changing world of education, there is no way basic education can be taken as a complete transformer of our society when the world at large is getting more complex and sophisticated'. Therefore, secondary education is critical as it provides additional knowledge, skills and attitudes crucial for enabling Malawians to cope with the complex and sophisticated socio-economic and political environment of the global village to which Malawi belongs. Specifically, secondary education is:

- a human right, and important for achieving gender equity
- important for improving the health and quality of life for individuals, families and communities
- important for the socio-economic and political development of the nation
- necessary for reaching the Millennium Development Goals (MDGs), Education For All (EFA) and for promoting Universal Primary Education (UPE)

Against this background, the Malawi Government through the Ministry of Education, Science and Technology has reviewed the secondary school curriculum with a view to improving its quality and relevance, and to align it with the primary curriculum which has since been reviewed and emphasises continuous assessment as a tool for learning.

The rationale for reviewing the secondary school curriculum is contained in the NESP (2008-2017), and PIF (2000). The documents clearly state that the purpose of secondary education is to provide students with the academic basis for gainful employment in formal, private and public sectors. They further state that secondary education will prepare students for further education according to their abilities and aptitudes. However, it is important to note that for the majority of the children in Malawi, secondary education is terminal. Consequently, the curriculum has put emphasis on practical skills that enable them to achieve self-employment.

It is also clear that a good secondary school curriculum enables a student to develop into an adult with sound intellectual, moral, physical, and emotional abilities. Therefore the curriculum needs to address the whole range of students' abilities and interests. In addition, it should aim at equipping the student to become an independent learner in order to promote personal, family, community and national development. The new curriculum has therefore been deliberately designed to achieve these important goals. The importance of this syllabus therefore, cannot be over-emphasised.

I would like to thank all those who were directly or indirectly involved in the preparation of the syllabuses. Key among the stakeholders are the Director and staff of the Department of Inspectorate and Advisory Services (DIAS) in the Ministry of Education, Science and Technology (MoEST), for facilitating the development of the syllabuses in collaboration with the Director and staff of the Malawi Institute of Education (MIE). I would also like to extend my gratitude to university colleges (both public and private), teachers from secondary schools (both public and private), members of different religious groups and officers representing special interest groups such as the Malawi Revenue Authority (MRA), Reserve Bank of Malawi (RBM), Malawi Bureau of Standards (MBS), Anti-Corruption Bureau (ACB) and Malawi Blood Transfusion Services (MBTS) for their valuable contribution to and participation in the preparation of these syllabuses.

Most of all, I would like to express my hope that teachers will implement this curriculum diligently and in the best interest of the students so that the goals for reviewing the curriculum are achieved.

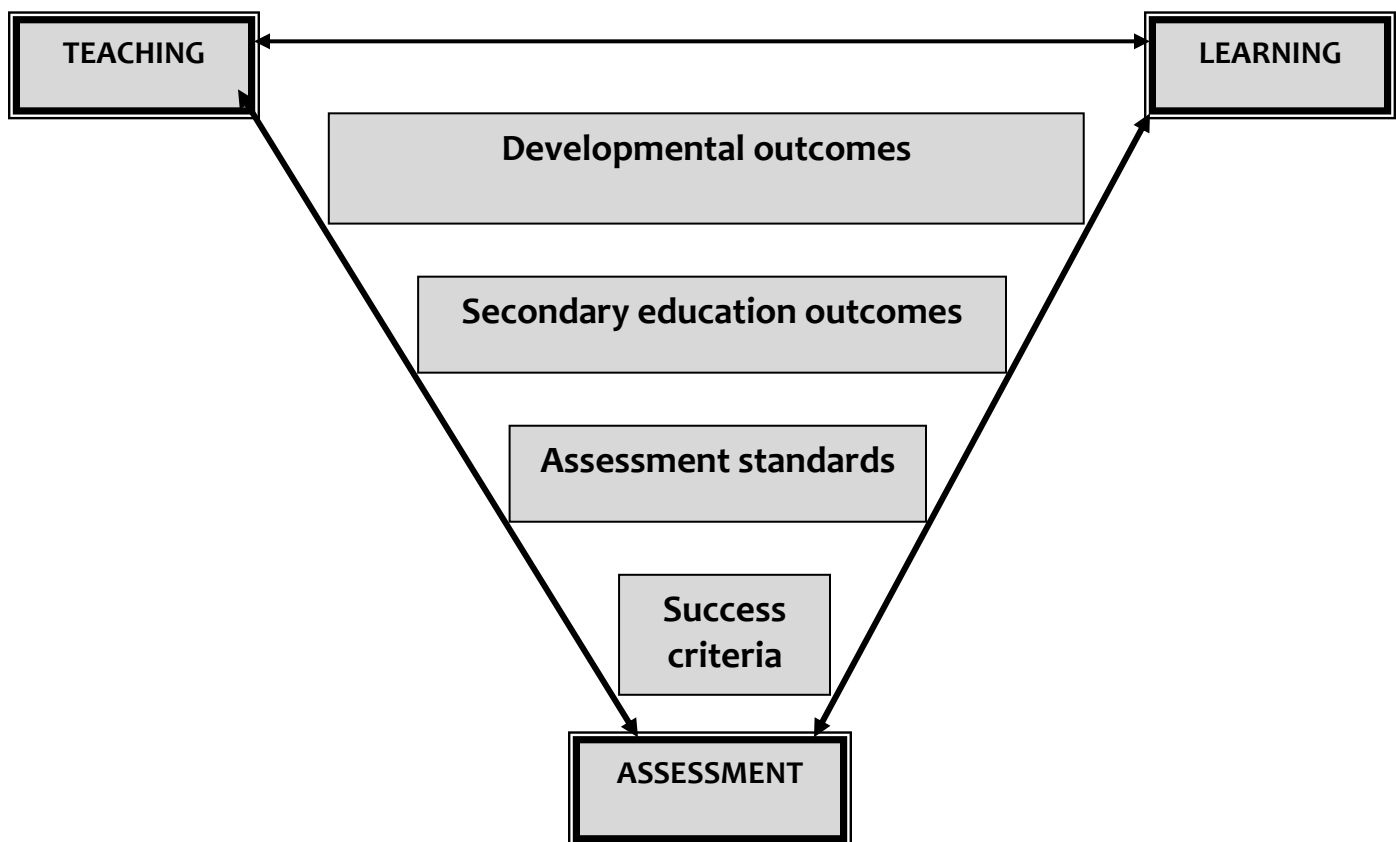
Dr McPhail Magwira
Secretary for Education Science and Technology (SEST)

The secondary school curriculum in Malawi

Among other reasons, the secondary school curriculum has been reviewed to align it with the primary school curriculum. This curriculum puts emphasis on student-centred teaching and learning approaches, including continuous assessment.

This curriculum focuses on student achievement. To achieve the outcomes, students must be introduced to new knowledge, skills, attitudes and values in the context of their existing knowledge, skills, attitudes and values so that they develop a deeper understanding as they learn and apply the knowledge. In this way, the process of learning is integral to the final product. The final products are the outcomes, that is, what students are expected to achieve in terms of knowledge, skills, attitudes and values, which must be clearly stated before teaching and learning begin. The achievements made at school, however, are only truly beneficial when the students transfer them to life beyond the school and view learning as a lifelong process. This is essential to keep pace with the changing social environment of home and work.

The figure below illustrates the structure and major elements of Malawi's secondary school curriculum, which are elaborated in the text below.



The developmental structure of the secondary school curriculum in Malawi

Developmental outcomes

The developmental outcomes are over-arching; they are what the student is expected to achieve by the end of the secondary school cycle both in and out of school. These outcomes apply to subject areas and they have been derived from the Constitution of the Republic of Malawi, Malawi Growth and Development Strategy (MGDS), National Education Sector Plan (NESP), Education Act and other education policy documents, including global policies and multilateral agreements to which Malawi is a signatory, as well as from the Secondary School Curriculum and Assessment Review (SSCAR). That is, students should be able to:

- 1 demonstrate appropriate moral and ethical behaviour in accordance with the accepted norms and values of the society
- 2 demonstrate local, regional, and international understanding
- 3 communicate competently, effectively, and relevantly in a variety of contexts, in an appropriate local or international language
- 4 apply mathematical concepts in socio-cultural, political, economic, environmental, scientific, and technological contexts to solve problems
- 5 apply scientific, technological, vocational, and managerial skills in a creative and innovative way to identify problems and develop appropriate solutions, so as to participate productively in society
- 6 demonstrate health-promoting behaviour in their personal lives as well as in their communities and the wider environment, with particular attention to prevalent diseases
- 7 appreciate and interact with the environment in a responsible and sustainable manner
- 8 apply the indigenous and non-indigenous knowledge and skills necessary for lifelong learning, personal advancement, employment, and the development of society
- 9 use Information and Communication Technology (ICT) responsibly and productively
- 10 demonstrate an understanding of the functioning of the economy and the contribution of agriculture and other sectors to national development
- 11 make use of entrepreneurial and vocational skills for personal and national development
- 12 apply research skills for problem-solving
- 13 demonstrate an understanding and appreciation of issues of human rights, democracy, gender, governance, and other emerging issues

Secondary education outcomes

The secondary education outcomes are categorised into seven sets of essential skills to be acquired by a secondary school graduate. The skills are:

- 1 citizenship skills
- 2 ethical and socio-cultural skills
- 3 economic development and environmental management skills
- 4 occupational and entrepreneurial skills
- 5 practical skills
- 6 creativity and resourcefulness
- 7 scientific and technological skills

Citizenship skills

- 1 demonstrate an understanding and appreciation of the symbols of nationhood
- 2 demonstrate a spirit of patriotism and national unity
- 3 apply decision-making skills necessary for participation in civic affairs
- 4 demonstrate a spirit of leadership and service
- 5 show respect for one's own and other people's rights and responsibilities
- 6 tolerate other people's attitudes and beliefs
- 7 demonstrate respect for the rule of law
- 8 understand characteristics of good governance
- 9 initiate and implement community development projects
- 10 demonstrate a sense of good neighbourliness
- 11 demonstrate a sense of national, regional and international understanding
- 12 demonstrate cooperative behaviour
- 13 demonstrate personal and social responsibility

Ethical and socio-cultural skills

- 14 demonstrate moral, spiritual and ethical attitudes and values
- 15 appreciate Malawi's diverse cultures and their respective practices
- 16 appreciate existing national institutions and cultural heritage
- 17 appreciate the value of the relationship between the individual and society
- 18 respect one's own and other people's cultures
- 19 identify beliefs which promote or retard national development
- 20 evaluate beliefs, taboos and superstitions in relation to national development
- 21 uphold beliefs which promote national development

Economic development and environmental management skills

- 22 understand Malawi's economy and economic structure
- 23 demonstrate entrepreneurial and/or vocational skills for formal or informal employment
- 24 exploit economic opportunities stemming from agriculture
- 25 demonstrate an interest in land husbandry, animal husbandry and aquaculture
- 26 apply appropriate agricultural practices and methods
- 27 acquire positive attitudes and skills, and apply them to the sustainable development of the natural and physical environment
- 28 understand the importance of diversified agriculture for Malawi's economy
- 29 understand the impact of technologies on economic productivity
- 30 apply relevant technologies to various economic activities
- 31 apply value addition practices to agricultural and environmental resource utilisation and management
- 32 appreciate Malawi's environmental resources
- 33 understand the impact of rapid population growth on natural resources and the delivery of social services
- 34 apply a variety of measures to conserve Malawi's natural resources
- 35 apply ICT skills to improve intellectual growth, personal enhancement and communication
- 36 demonstrate the ability to adapt to climate change and mitigate its impact on the economy and environment

- 37 appreciate the importance of energy in economic development
- 38 understand the importance of diversifying the economy through sectors such as tourism, mining and manufacturing

Occupational and entrepreneurial skills

- 39 demonstrate the spirit of self-reliance through vocational and entrepreneurial activities
- 40 apply appropriate vocational, occupational and entrepreneurial skills to individual and national advancement
- 41 demonstrate effective communication skills for the transfer of occupational and entrepreneurial knowledge, skills, attitudes and values
- 42 apply the principles of science and technology, entrepreneurship and management to promote active and productive participation in the society
- 43 demonstrate creativity and innovation for the benefit of the individual, community and the nation as a whole
- 44 demonstrate an understanding of indigenous and non-indigenous knowledge, skills, attitudes and values, and apply them to personal intellectual growth and national development
- 45 use vocational, occupational and entrepreneurial skills for the creation of economic opportunities in agriculture and other sectors

Practical skills

- 46 acquire entrepreneurial skills related to agriculture, commerce and industry
- 47 apply appropriate skills to agricultural, commercial and industrial production
- 48 demonstrate positive attitudes to manual work
- 49 demonstrate excellence in any kind of workmanship
- 50 demonstrate sporting ability and sportsmanship
- 51 demonstrate the ability to use creative and innovative artistic talents for self-employment

Creativity and resourcefulness

- 52 demonstrate a spirit of inquiry and creative, critical and lateral thinking
- 53 use problem-solving techniques to solve practical problems
- 54 demonstrate an imaginative and creative mind
- 55 exploit creative potential
- 56 understand personal strengths and weaknesses and use strengths to promote healthy self-esteem
- 57 maximise the use of available resources

Scientific and technological skills

- 58 apply appropriate scientific, technological and vocational skills to improve economic productivity
- 59 apply relevant innovations in science and technology
- 60 demonstrate a capacity to utilise appropriate technology
- 61 demonstrate basic research skills

Rationale for agriculture

The economy of Malawi is predominantly agricultural, with about 90% of the population living in rural areas. The sector is the single largest employer and generates around 80 % of the country's foreign currency. The majority of secondary school graduates therefore end up participating directly or indirectly in the agricultural sector.

Agriculture in secondary schools equips the student with the requisite knowledge and skills, to apply scientific, analytical, research, technological, entrepreneurial, critical thinking, vocational and managerial skills in a creative and innovative way. This helps the students to identify problems and develop their appropriate solutions for personal and national development. It also enables secondary school graduates to apply the knowledge, skills and attitudes for environmental conservation, sustainable crop and animal production, farm business management, value addition and efficient marketing of products to maximize profits.

Further, the subject provides a foundation for those proceeding to tertiary levels of education and related fields.

Core elements and their outcomes

Agriculture and the environment

The students will be able to demonstrate an awareness of the environmental and climatic factors which influence agricultural production, how to modify them to reduce risks in order to maximize yield, and conserve them for sustainability.

Agricultural research and technology

The students will be able to apply research skills and apply improved agricultural technology to increase farm productivity.

Agricultural economics and farm business management

The students will be able to appreciate how concepts and principles of agricultural economics and farm business management increase profitability of agricultural enterprises.

Crop production

The students will be able to use scientific knowledge and skills of growing crops using recommended husbandry practices.

Livestock production

The students will be able to use scientific knowledge and skills of rearing farm animals profitably using appropriate husbandry practices.

Scope and sequence chart for Forms 1 and 2

Core element	Form 1	Form 2
Agriculture and the environment	Natural resources that influence agricultural production • definition of natural resources • importance of the natural resources • depletion of natural resources • conservation of the natural resources • environmental factors influencing agricultural production • effects of rapid population growth on the natural resources	Soil composition Soil profile • characteristics of soil profile • effects of soil profile on crop production Soil fertility • maintenance of soil fertility • classification of fertilizers • advantages and disadvantages of organic and inorganic fertilizers Agriculture and climate change • causes • effects
Agricultural research and technology	Agricultural research • agricultural problems requiring research • scientific approach to agricultural research Agricultural technology • meaning • importance of agricultural technology Agricultural technology • types of farm structures • safety rules to be observed when utilizing farm structures	Agricultural technology • Irrigation - importance of irrigation - types of irrigation system - advantages and limitations of each irrigation system - factors affecting frequency and time of irrigation - factors to consider when selecting an irrigation unit - procedure to be followed when establishing an irrigation unit - maintenance of irrigation systems

Core element	Form 1	Form 2
		Agricultural research • experimental designs • laying out experiments • conducting an agricultural research • importance of report writing • format for report writing
Agricultural economics and farm business management	Farm business management concepts Factors of agricultural production • main factors of agricultural production • other factors of agricultural production • solutions to challenges associated with factors of agricultural production Agricultural marketing • marketing forces • marketing functions • problems associated with marketing agricultural products and possible solutions Agro-based industries • agro-based industries in Malawi • role of agro-based industries in supporting the growing population Food distribution • food security and food for self-sufficiency • relationship between family size and food security	Gender and agricultural production • gender roles in agricultural production • the impact of gender roles in agricultural production • gender equity in agricultural production • role of women in decision making in agricultural production Farming systems • types of farming systems • importance of farming systems Financing agricultural enterprises • sources of finance • calculating interest on borrowed money

Core element	Form 1	Form 2
Crop production	Plant nutrients • essential plant nutrients • classes of essential plant nutrients • role of essential plant nutrients • depletion of essential plant nutrients from the soil • deficiency signs of essential plant nutrients • sources of essential plant nutrients Vegetable production • importance of vegetables • types of vegetables • advantages of indigenous vegetables • site selection for vegetable growing • construction of garden fence • husbandry practices in vegetable growing	Maize production • qualities of good seed • husbandry practices Groundnuts production • qualities of good seed • husbandry practices
Livestock production	Classes of livestock • digestive systems of ruminants and non-ruminants • functions of different parts of digestive systems of ruminants and non-ruminants Chicken production • breeds and breeding • housing • feeds and feeding • disease and parasite control • predators of chickens	Pig production • breeds and breeding • management systems • housing • feeds and feeding • disease and parasite control

Scope and sequence chart for Forms 3 and 4

Core element	Form 3	Form 4
Agriculture and the environment	Physical properties of soil • soil texture • soil structure • soil porosity • soil colour • soil consistency • soil temperature • soil depth Effects of physical properties of soil on crop production Chemical properties of soil • soil pH • Cation Exchange Capacity of soil • nutrient status of soil • salinity • factors affecting chemical properties of the soil • effects of chemical properties of soil on crop production	Soil degradation • forms of soil degradation • causes of soil degradation • effects of soil degradation on crop production • relationship between population growth and soil degradation • ways of controlling soil degradation Agriculture and climate change: mitigation measures to climate change • conservation agriculture • agro-forestry • re-afforestation • integrated forest, crop, livestock and fish systems Land drainage • importance of land drainage • methods of land drainage
Agricultural research and technology	Agricultural development agencies and services • agricultural development agencies in Malawi • services offered by the agricultural development agencies • importance of agricultural development agencies to the growing population	Farm mechanization • types of farm machinery • factors to consider when mechanizing a farm • advantages of farm mechanization • limitations of farm mechanization • maintenance of farm machinery • safety measures when using farm machinery

Core element	Form 3	Form 4
		Farm power • sources of power on a farm • ways of improving output of each source of farm power • advantages and disadvantages of each source of farm power Agricultural technology and gender • gender biases on agricultural technology • causes of gender biases in agricultural technology • implications of gender biased application of agricultural technology • ways of dealing with gender biases in agricultural technology Improved farming technology • improved farming technology and food supply • improved farming technology and food security
Agricultural economics and farm business management	Farm records • types • reasons for keeping records Farm budgets • types of farm budget • budgeting for an agricultural enterprise	Agricultural marketing and trading • difference between marketing and trading • marketing channels and marketing agencies • roles of marketing channels and agencies

Core element	Form 3	Form 4
	Economic principles that aid in farm business decision making Enterprise combination • selection of an enterprise combinations • types of enterprise combinations Agricultural cooperatives • importance of agricultural cooperatives • formation of agricultural cooperatives • challenges of running agricultural cooperatives • solutions to challenges faced in running agricultural cooperatives	• marketing costs and marketing margins • effects of population distribution on marketing • trading of agricultural commodities • improving trading of agricultural commodities Price elasticity of demand and supply • calculation of price elasticity of demand and supply
Crop production	Vegetative planting materials • parts of vegetative planting materials • advantages of using vegetative planting materials • disadvantages of using vegetative planting materials Cropping systems • types of cropping systems • activities of the cropping systems • advantages of different cropping systems • disadvantages of different cropping systems Mushroom production • species of mushroom • importance of mushrooms • husbandry practices for mushroom production • site selection for mushroom production • constructing of incubation and production sheds for mushroom	Crop improvement • aims of crop improvement • methods of crop improvement Crop processing • processing of various crops for storage Pasture production • importance of pasture production • types of pastures • methods of pasture establishment • pasture management (methods of pasture management, grazing systems and pasture conservation methods) • factors affecting the quality of conserved pasture Mango production • importance of mangoes • mango varieties • site selection for mango production

Core element	Form 3	Form 4
	• preparing substrate for mushroom production • mushroom fruiting management • mushroom harvesting	• site preparation for mangoes • husbandry practices for growing mangoes
Livestock production	Feeds and feeding • classes of livestock feeds • feed nutrients • functions of feed nutrients • sources of feed nutrients • importance of feeding livestock • factors to consider when feeding livestock Sheep and goat production • breeds of sheep and goats • housing sheep and goats • feeding sheep and goats • diseases of sheep and goats • parasites of sheep and goats	Cattle production • breeds of cattle (dairy and beef) • managing cattle Reproductive systems in cattle and poultry Age of puberty for different animals Oestrus cycle for cow Gestation period for different animals Reproduction in cattle Livestock improvement

Teaching syllabus for Forms 3 and 4

Form 3

Core element: Agriculture and the environment

Outcome: The students will be able to demonstrate an awareness of the environmental factors which influence agricultural production, how to modify them to reduce risks in order to maximise yield, and conserve them for sustainability.

Assessment standard	Success criteria	Theme/ Topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of soil properties	The students must be able to: list the physical properties of soil	Physical properties of soil	- brainstorming the physical properties of soil - listing the physical properties of soil - texture - structure - colour - consistency - porosity - temperature - depth - summarizing the physical properties of soil	- brainstorming - discussion - question and answer - oral questions - research	- students' experiences - libraries - charts - ICT

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	2 describe the physical properties of soil		• discussing physical properties of soil • visiting the school garden to observe different physical properties of soil • reporting findings	• discussion • question and answer • presentation • excursion • observation • explanation	• local environment • students' experiences • charts • libraries • ICT
	3 determine the physical properties of soil		• visiting the local environment to observe different types of soils • collecting samples of different types of soils • demonstrate how to determine physical properties of soil – texture – colour – consistency – porosity – temperature – depth • reporting findings • summarizing how to determine physical properties of soil	• discussion • presentation • excursion • demonstration • observation • explanation • questions and answers • report writing • oral questions • practical exercises	• local environment • students' experiences • charts • libraries • written reports • soil samples • equipment • checklists

Assessment standard	Success criteria	Theme/ Topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of soil structure	The students must be able to: 1 identify types of soil structure 2 describe how soil structure can be destroyed	Soil structure	- researching on types of soil structure - reporting the findings - discussing types of soil structure - block - cuboidal - columnar - prismatic - platy - granular - crumb - single grain - researching on how soil structure can be destroyed - reporting findings - discussing how soil structure can be destroyed	- discussion - presentation - explanation - library search - web search - quizzes - question and answer - report writing - research - presentation - discussion - quizzes	- students' experiences - libraries - written reports - charts - pictures - soil samples - models - ICT - students' experiences - libraries - written reports - pictures - charts - local environment

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 describe methods of maintaining and improving soil structure 4 explain the effects of physical properties of soil on crop production		• brainstorming methods of maintaining and improving soil structure • discussing methods of maintaining and improving soil structure • demonstrating how to conduct experiments on maintaining and improving soil structure • conducting experiments on maintaining and improving soil structure • reporting findings • brainstorming effects of physical properties of soil on crop production • discussing the effects of physical properties of soil on crop production - texture - structure - colour - consistency - porosity - temperature - depth • summarizing the effects of physical properties of soil on crop production	• brainstorming • discussions • demonstration • practical exercise • reporting • explanation • brainstorming • discussion • explanation • oral questions • library search • web search	• local environment • students' experiences • charts • equipment • soil samples • students' experiences • charts • libraries • ICT

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of the chemical properties of soil	The students must be able to: list the chemical properties of soil	Chemical properties of soil	- brainstorming the chemical properties of soil - listing chemical properties of soil in terms of - nutrient status - pH - salinity - cation exchange capacity - summarizing the chemical properties of soil	- brainstorming - explanation - oral tests - written assessment - library search - web search	- students' experiences - internet - libraries - ICT

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	2 discuss the chemical properties of soil		- collecting soil samples from the local environment - discussing the procedure for determining the chemical properties of soil - soil pH - salinity - demonstrating the procedure for determining the chemical properties of soil - soil pH - salinity - determining the chemical properties of soil - reporting findings - summarizing the procedure for testing soil pH and salinity	- excursion - demonstration - practical exercises - discussion - presentation - quizzes	- local environment - soil samples - soil PH equipment - students' experiences - salinity equipment - evaporating basins/tins - source of heat
	3 explain factors affecting soil pH		- brainstorming the meaning of the term <i>soil pH</i> - discussing the meaning of the term <i>soil pH</i> - researching on factors affecting soil pH - reporting findings	- brainstorming - researching - discussion - presentation - quizzes - written exercise	- libraries - students' experiences - charts - reports - ICT

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	4 modify soil pH		• suggesting ways of modifying soil pH • discussing ways of modifying soil pH - application of agricultural lime - application of manure - application of inorganic fertilizers • collecting resources required for modifying soil pH - agricultural lime - manure - inorganic fertilizers • collecting soil samples from the local environment • discussing the procedure for modifying soil pH • demonstrating how to modify soil pH • modifying soil pH	• excursion • demonstration • experimentation • discussion • presentation • quizzes • explanation • teachers' observation on students ability to modify pH	• local environment • soil samples • soil pH equipment • students' experiences • materials for modifying soil pH
	5 explain the factors that affect the nutrient status of the soil		• brainstorming factors that influence the nutrient status of the soil • visiting the local environment to observe factors affecting nutrient the status of soil • discussing the factors that influence the nutrient status of the soil	• brainstorming • report writing • excursion • discussion • presentation • quizzes	• local environment • resource persons • students' experiences • charts

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	6 explain effects of chemical properties of soil on crop production		- parent material of soil - soil pH - leaching of basic nutrients - soil erosion - nutrient uptake by plants - crop removal - method of cultivation • recording observations • reporting observations • brainstorming effects of chemical properties of soil on crop production • discussing the effects of chemical properties of soil on crop production - nutrient status - soil pH - salinity - cation exchange capacity • summarizing the effects of chemical properties of soil on crop production	• explanation • library search • web search • brainstorming • discussion • explanation • oral questions • library search • web search	• libraries • ICT • libraries • students' experiences • charts • ICT

Core element: Agricultural research and technology

Outcome: The students will be able to apply research skills, use and maintain farm machinery, apply improved agricultural technology to increase farm productivity.

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when students are able to: demonstrate an understanding of agricultural development agencies and their services in Malawi	The students must be able to: 1 identify agricultural development agencies and their services in Malawi	Agricultural development agencies and their services	- brainstorming development services required in agriculture – research – marketing – infrastructure – extension – processing – production – credit - identifying agencies offering agricultural services - visiting a nearby agricultural agency to observe services offered - evaluating the availability of agricultural development and the services provided in an area - reporting findings	- brainstorming - excursion - presentation - discussion - question and answer - written tests - oral assessment - matching tests	- students' experiences - resource persons - rubrics - checklists - agricultural development agencies - test items

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	2 describe the services offered by agricultural development agencies 3 relate the importance of agricultural development agencies on a growing population		- brainstorming the services offered by the agricultural development agencies - discussing the services offered by the agricultural development agencies - summarizing the services offered by the agricultural development agencies - brainstorming the importance of agricultural development agencies to a growing population - discussing the importance of agricultural development agencies to a growing population - summarizing the importance of agricultural development agencies to a growing population	- brainstorming - discussion - question and answers - written tests - oral assessment - explanation - research - brainstorming - discussion - question and answer - written tests - oral assessment - explanation - research	- students' experiences - resource persons - charts - pictures - libraries - ICT - students' experiences - resource persons - charts - pictures - libraries - ICT

Core element: Agricultural economics and farm business management

Outcome: The students will be able to appreciate how concepts and principles of agricultural economics and farm business management increase profitability of agricultural enterprises.

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of farm records	The students must be able to: 1 describe types of farm records 2 state the reasons for keeping farm records	Farm records	- brainstorming examples of farm records - categorizing farm records – inventory records – production records – financial records - researching on reasons for keeping farm records - discussing reasons for keeping farm records - reporting findings	- brainstorming - question and answer - library search - web search - matching exercises - case study - library search - brainstorming - discussion - questions and answers - written exercises	- charts - libraries - ICT - students' experiences - samples of farm records - students experiences - resource persons - samples of farm records

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 demonstrate keeping farm records		- visiting a nearby farm to observe how records are kept - demonstrating layout of different types of farm records - keeping records of crop and livestock enterprises	- research - demonstration - practice - project - excursion - written exercises	- students' experiences - charts - local environment - resource persons - checklists - libraries

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of farm budgets	The students must be able to: - define the term <i>budgeting</i> - state types of farm budgets	Budgeting	- brainstorming the meaning of the term <i>budgeting</i> - discussing the meaning of the term <i>budgeting</i> - summarizing the meaning of the term <i>budgeting</i> - brainstorming types of farm budgets - discussing types of farm budgets - partial budgets - break even budgets - complete budgets - visiting nearby farms to see different types of budgets - reporting findings	- brainstorming - discussion - question and answers - multiple choice questions - matching test - brainstorming - discussion - field visit - report writing - question and answer	- students' experiences - samples of agricultural budgets - test items - written reports - charts - local environment - written exercises

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 discuss how to budget for an agricultural enterprise		• brainstorming activities involved in budgeting for an agricultural enterprise • discussing activities involved in budgeting for an agricultural enterprise • collecting data from agricultural enterprises • preparing budgets using collected figures • summarizing the steps involved in budgeting	• brainstorming • discussion • practical exercises • question and answers • written exercises • oral exercises	• students' experiences • resource persons • local environment • charts • records

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: • demonstrate an understanding of economic principles that assist in farm business decision making	The students must be able to: 1 state economic principles in farm business decision making	Farm business decision making	• researching on economic principles farmers use in deciding on: – what to produce – how to produce – how much to produce – when to produce – where to buy and sell • reporting findings • summarizing the economic principles that farmers use in decision making – opportunity cost – comparative advantage – substitution of inputs – diminishing marginal returns	• research • presentation • discussion • explanation • oral questions	• ICT • chart paper • students experiences • local environment

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	2 explain how each of the economic principles help in decision making in farm business 3 draw a graph to illustrate the law of diminishing marginal return 4 interpret the graph as a farm business management tool		- brainstorming how each of the economic principles helps in decision making in farm business - discussing how each of the economic principles helps in decision making in farm business - summarizing how each of the economic principles helps in decision making in farm business - opportunity cost - collecting data on level of input and the corresponding yield for a given enterprise - tabulating collected data - drawing a graph using tabulated data - discussing what a graph shows - identifying the level of input that gives the highest marginal returns	- brainstorming - discussion - explanation - question and answer - written exercises - explanation - survey - discussion - teacher observation - drawing - discussion - explanation - question and answer - written test	- students' experiences - local environment - graph paper - students' experiences - local environment - chart paper - checklists - students' experiences - test items - chart paper

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of enterprise combinations	The students must be able to: - outline factors to be considered when selecting enterprise combinations - explain types of enterprise combinations	Enterprise combinations	- brainstorming factors to be considered when selecting enterprise combinations - explaining factors to be considered when selecting enterprise combinations - visiting nearby farms to investigate enterprise combinations - discussing the types of enterprise combination with examples - complementary - supplementary - competitive - summarizing types of enterprise combinations - choosing appropriate enterprise combinations	- brainstorming - explanation - question and answer - oral questions - surveys - excursions - excursion - explanation - discussion - question and answer - project - teacher observation	- students' experiences - charts - libraries search - local environment - students' experiences - checklists

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of agricultural cooperatives	The students must be able to: 1 explain the importance of agricultural cooperatives 2 identify principles for agricultural cooperative formation	Agricultural cooperatives	- brainstorming the meaning of the term <i>agricultural cooperative</i> - discussing the meaning of the term <i>agricultural cooperative</i> - researching on the importance of agricultural cooperatives - visiting an agricultural cooperative to observe its activities - reporting findings - discussing the importance of agricultural cooperatives - summarizing the importance of agricultural cooperatives - visiting nearby agricultural cooperatives to research on their formation - reporting findings - summarizing the formation of agricultural cooperatives	- brainstorming - discussion - excursion - question and answer - research - short essays - presentation - excursion - question and answer - explanation	- resource persons - students' experiences - written reports - local environment - ICT - libraries - students experiences - charts - resource persons

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 explain challenges of running agricultural cooperatives 4 describe solutions to the challenges faced in running agricultural cooperatives		- brainstorming the challenges of running agricultural cooperatives - discussing the challenges of running agricultural cooperatives - suggesting solutions to the challenges faced in running agricultural cooperatives - discussing solutions to the challenges faced in running agricultural cooperatives - summarizing solutions to the challenges faced in running agricultural cooperatives	- brainstorming - discussion - question and answer - explanation - research - brainstorming - discussion - question and answers - explanation - research	- students' experiences - resource persons - libraries - local community - ICT - Students' experiences - resource persons - libraries - ICT - local community

Core element: Crop production

Outcome: The students will be able to use scientific knowledge and skills of growing selected crops using recommended husbandry practices.

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of propagating crops vegetatively	The students must be able to: list vegetative planting materials	Vegetative planting materials	- recalling types of reproduction - brainstorming examples of vegetative planting materials - observing samples of vegetative planting materials - outlining examples of vegetative planting materials - stems - leaves - suckers - tubers - bulbs - corms - rhizomes - runners	- brainstorming - discussion - written exercises - teacher observation	- students' experiences - local environment - samples of vegetative planting materials

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	2 label parts of vegetative planting materials 3 discuss the advantages of vegetative planting materials		- identifying parts of vegetative planting materials - drawing diagrams of vegetative planting materials - labelling parts of vegetative planting materials - summarizing parts of vegetative planting materials - brainstorming the advantages of vegetative planting materials - discussing the advantages of vegetative planting materials - reporting findings - listing the advantages of vegetative planting materials	- brainstorming - research - demonstration - presentation - drawing - written exercises - observation - practical assessment - teachers observation - brainstorming - discussion - presentation - written exercises - research	- checklists - students' experiences - local environment - charts - pictures - vegetate planting materials - libraries - ICT - checklists - students' experiences - Charts - libraries - ICT
Assessment	Success	Theme/	Suggested teaching and learning	Suggested teaching,	Suggested teaching

standard	criteria	topic	activities	learning and assessment methods	and learning resources
	4 discuss the disadvantages of vegetative planting materials		- brainstorming the disadvantages of vegetative planting materials - discussing the disadvantages of vegetative planting materials - reporting findings - listing the disadvantages of vegetative planting materials	- brainstorming - discussion - presentation - written assessment - oral assessment - research	- students' experiences - charts - libraries - ICT

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of cropping systems	The students must be able to: 1 outline cropping systems	Cropping systems	- brainstorming the meaning of the term 'cropping systems' - discussing the meaning of the term 'cropping systems' - researching on cropping systems - outlining cropping systems - monoculture - monocropping - continuous cropping - mixed cropping - crop rotation - bush fallowing - shifting cultivation - organic farming - agro forestry	- brainstorming - discussion - explanation - question and answer - oral tests - written assessment	- students' experiences - libraries - ICT - charts - pictures

Assessment	Success	Theme/	Suggested teaching and learning	Suggested teaching,	Suggested teaching
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standard	criteria	topic	activities	learning and assessment methods	and learning resources
	2 analyze activities in the cropping systems		- researching on activities for each cropping system - reporting their findings - discussing activities of each cropping system - summarizing the activities of each cropping system	- research - presentation - discussion - explanation - question and answer - oral tests - written assessment - short essays	- students' experiences - libraries - ICT - resource persons - reports - charts - pictures
	3 explain the advantages of cropping systems		- brainstorming the advantages of different cropping systems - discussing the advantages of different cropping systems	- brainstorming - discussion - explanation - question and answers - oral tests - written assessment	- students' experiences - libraries - ICT - resource persons - charts - pictures
Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources

	4 explain the disadvantages of different cropping systems 5 discuss cropping systems which can promote crop yields		• brainstorming the disadvantages of different cropping systems • discussing the disadvantages of different cropping systems • evaluating cropping systems which can promote crop yields • choosing cropping systems which can promote crop yields • summarizing cropping systems which can promote crop yields	• brainstorming • discussion • explanation • question and answers • oral tests • written assessment • discussion • explanation • question and answer • oral tests • written tests • research	• students' experiences • libraries • ICT • resource persons • charts • pictures • students' experiences • libraries • ICT • charts
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this	The students				

when the students are able to : demonstrate an understanding of growing mushrooms	must be able to : 1 describe mushrooms	Mushroom production	- brainstorming parts of a mushroom - discussing parts of a mushroom - drawing a well-labelled diagram of a mushroom - consolidating parts of a mushroom	- drawing - discussing - question and answer - brainstorming - research - library search - excursion	- local environment - students' experiences - ICT - charts - pictures - samples of mushroom
	2 identify cultivated species of mushroom		- visiting a nearby farm to identify cultivated species of mushroom - identifying cultivated species of mushroom from pictures - listing cultivated mushroom species - choosing suitable species for mushroom production for your area - summarizing species of mushroom	- excursion - discussion - brainstorm - question and answer	- local environment - students' experiences - samples of mushroom - pictures - ICT

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 explain the importance of		brainstorming the importance of mushrooms	- library research - presentation	students' experiences

	mushrooms		• discussing the importance of mushrooms • presenting the importance of mushrooms • brainstorming the husbandry practices for mushroom production • listing husbandry practices for mushroom production – species selection – site selection – construction of incubation and production shed – substrate preparation – spawn source (buying or preparation) – substrate treatment – mushroom seeding – mushroom fruiting management – harvesting	• web research • explanation • written exercise • oral questions • brainstorming • discussion • excursion • oral/written assessment • research	• ICT • samples of medicines made from mushroom • students' experiences • libraries • ICT • local environment
4	list the husbandry practices for mushroom production				

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	5 describe a suitable site for mushroom		• brainstorming the qualities of a suitable site for mushroom production • discussing the qualities of a suitable site	• brainstorming • discussion • excursion	• local environment • students'

	production		for mushroom production - visiting possible site for mushroom production - reporting the findings - selecting suitable site for mushroom production - summarizing qualities of a suitable site for mushroom production	- presentation - question and answer - practical work	experiences libraries
	6 discuss how to construct incubation and production sheds for the mushroom production		- brainstorming suitable materials for construction of incubation and production sheds for mushroom - listing suitable materials for construction of incubation and production sheds for mushroom - assembling materials for construction of incubation and production sheds for mushroom - demonstrating the construction of incubation and production sheds for mushroom - constructing incubation and production sheds for mushroom	- brainstorming - demonstration - practical work - teachers observation written/oral assessment	- construction materials - local environment - students' experiences - ICT - checklists - rubrics - question items
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	7 prepare substrate for mushroom production		- brainstorming the meaning of the term <i>substrate</i> in mushroom production - identify materials suitable for	- brainstorming - demonstration - excursion - practical	- local environment - students' experiences

	8	treat substrate for mushroom production	- substrate preparation - collecting materials for substrate preparation - demonstrating the preparation of substrate - preparing substrate for mushroom production - researching different ways of treating substrate for mushroom production - reporting findings - summarizing substrate treatment - demonstrating treatment of substrate - treating substrate	- teachers' observation of students' participation in preparing substrate - research - presentation - demonstration - explanation - practical - teachers' observation of students' participation in treating the substrate	- materials (maize stalks, maize sheaths) - rubrics - checklists - local environment - libraries - ICT - students' experiences - materials for treating substrates
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	9	discuss how to seed the substrate for mushroom production	- researching on the seeding of substrate for mushroom production - reporting findings - summarizing the seeding of substrate for mushroom production - demonstrating how to seed	- research - presentation - demonstration - explanation - practical - written/oral	- local environment - libraries - ICT - students' experiences - spawn

			- substrate for mushroom production - seeding substrate for mushroom production	- assessment - teachers' observation of students' participation in seeding substrate	- methyalted spirit - rubrics - checklists
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	10 discuss management of mushroom fruiting		- researching on the management practices of mushroom fruiting - reporting the findings - discussing the management of mushroom fruiting - monitoring spawn running, - monitoring colour change - hanging in production sheds	- research - presentation - explanation - discussion - demonstration - practical - teacher observation	- libraries - ICT - local environment - mushroom sheds - materials for managing mushroom fruiting - students'

	11 discuss mushroom harvesting		- watering • demonstrating the management of mushroom fruiting • managing mushroom fruiting • brainstorming factors to consider when harvesting mushroom • discussing factors to consider when harvesting mushrooms • consolidating factors to consider when harvesting mushrooms • demonstrating how to harvest mushrooms • harvesting mushroom	• brainstorming • discussing • explanation • demonstration • practical • teachers' observation of students' participation in harvesting mushroom	- experiences • resource persons • materials for harvesting mushroom • mushroom • students' experiences • resource persons
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Core element: Livestock production

Outcome: The students will be able to use scientific knowledge and skills of rearing farm animals profitably using appropriate husbandry practices.

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: • demonstrate an understanding of livestock feeding.	The students must be able to: 1 distinguish classes of livestock feeds	Livestock feeds and feeding - types - feed nutrients	• observing samples of livestock feeds • discussing types of livestock feeds provided	• brainstorming • discussion • presentation • practical work	• livestock feeds samples • students' experiences

		– functions of livestock feeds – sources – importance – what to consider when feeding livestock	• classifying the provided feeds – concentrates – roughages • distinguish classes of feeds • consolidating classes of livestock feeds	• oral test • written tests • matching test • teacher observation	• pictures • charts • checklists • rubrics
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	2 list nutrients in livestock feeds		• brainstorming nutrients in livestock feeds • discussing nutrients in livestock feeds • naming nutrients in livestock feeds – water – carbohydrates – fats and oils – proteins – vitamins – minerals	• brainstorming • discussion • presentation • oral test • written tests • research	• students' experiences • charts • ICT • libraries

	3 explain functions of nutrients in livestock feeds 4 identify sources of nutrients in livestock feeds		- researching on the functions nutrients in livestock feeds - discussing the functions of nutrients in livestock feeds - brainstorming sources of nutrients in livestock feeds - discussing sources of nutrients in livestock feeds reporting findings - summarizing sources of nutrients in livestock feeds	- discussion - presentation - oral assessment - written assessment - matching tests - brainstorming - discussion - presentation - oral tests - written tests	- students' experiences - pictures - charts - checklists - rubrics - students' experiences - charts - pictures - feed samples
Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	5 explain the importance of feeding livestock 6 explain factors to consider when feeding livestock		- researching on the importance of feeding livestock - discussing the importance of feeding livestock - brainstorming factors to consider when feeding livestock - listing factors to consider when feeding livestock - discussing factors to	- discussion - presentation - oral tests - written tests - role-play - research - brainstorming - discussion - presentation - oral tests - written tests	- students experiences - pictures - charts - libraries - ICT - resource persons - livestock feeds - learners' experiences - pictures - charts - libraries

			consider when feeding livestock		• ICT • resource persons
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: • demonstrate an understanding of sheep and goats production	The students must be able to: 1 list breeds of sheep and goats 2 select an appropriate breed of sheep and	Sheep and goats production – breeds – housing – feeding	• researching on breeds of sheep and goats • recording findings • listing breeds of sheep and goats • classifying breeds of sheep and goats according to their use eg meat, milk and wool • visiting nearby village to observe characteristics of different breeds of sheep and goats • identifying different breeds of	• brainstorming • discussion • presentation • oral tests • written assessment • matching tests • research	• students' experiences • pictures • library search • web search • resource persons

	goats for their area		sheep and goats - researching on appropriate breeds of sheep and goats for the area - reporting findings - choosing appropriate breeds of sheep and goats based on their use		
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 identify a suitable sheep and goats house 4 construct an appropriate house for sheep and goats		- brainstorming the characteristics of a good sheep and goats house - discussing the characteristics of a good sheep and goats house - visiting a nearby village to observe sheep and goats houses - reporting findings - choosing an appropriate sheep and goats house - brainstorming materials required for constructing sheep and goats house - collecting appropriate materials for construction of sheep and goats house - discussing the procedure for constructing sheep and goats	- discussion - presentation - oral tests - written assessment - teacher' observation of students ability to identify appropriate breeds based on their use	- nearby livestock unit - students' experiences - pictures - charts - libraries - ICT - resource persons

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Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of diseases of sheep and goats	The students must be able to : 1 identify diseases of sheep and goats 2 explain how to control diseases of sheep and goats	Sheep and goats production – diseases – parasites	- researching on the diseases of sheep and goats - listing diseases of sheep and goats - describing signs and symptoms of diseases of sheep and goats - role-playing signs and symptoms of diseases of sheep and goats - recording the signs and symptoms of diseases of sheep and goats - brainstorming control measures of diseases of sheep and goats - discussing the control measures of diseases of sheep and goats - observing a resource person controlling diseases of sheep and	- brainstorming - discussion - presentation - oral tests - written tests - matching tests - teachers' observation of students ability to identify diseases - role-play - brainstorming - discussion - presentation - practical - oral tests - written tests	- students' experiences - pictures - charts - libraries - ICT - resource persons - pictures - charts - libraries - resource person - drugs - equipment

			goats summarizing the control measures of diseases of sheep and goats	demonstration	- ICT - nearby livestock units
Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methodologies	Suggested teaching and learning resources
	3 identify parasites of sheep and goats 4 explain how to control parasites of sheep and goats		- researching on the parasites of sheep and goats - listing parasites of sheep and goats - describing the signs of parasites in sheep and goats - discussing the signs of parasite attack in sheep and goats - recording the signs of parasites of sheep and goats - brainstorming control measures of parasites in sheep and goats production - discussing the control measures of parasites in sheep and goats production - observing a resource person controlling parasites in sheep and goats production - summarizing the control measures of parasites in sheep and goats production	- brainstorming - discussion - presentation - oral tests - written tests - role play - teacher's observation of students' ability to identify parasites - role-playing - brainstorming - discussion - presentation - practical work - oral test - written assessment - demonstration	- students' experiences - pictures - charts - libraries - resource persons - samples of parasites - nearby livestock units - students' experiences - pictures - charts - libraries - ICT - resource persons - drugs - equipment

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Form 4

Core element: Agriculture and the environment

Outcome: The students will be able to demonstrate an awareness of the environmental factors which influence agricultural production, how to modify them to reduce risks in order to maximise yield, and conserve them for sustainability.

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching learning and assessment resources
We will know this when the students are able to: demonstrate an understanding of soil degradation	The students must be able to: describe forms of soil degradation	Soil degradation	- brainstorming the meaning of soil degradation - discuss the meaning of soil degradation - visiting degraded areas - observing and recording forms of soil degradation - brainstorming forms of soil degradation - discussing forms of soil degradation - physical - chemical - summarizing forms of soil degradation	- excursion - observation - presentation - discussion - explanation - question and answer - oral questions - written questions	- local environment - students' experiences - resource persons - charts - pictures - checklists

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	2 explain the causes of soil degradation 3 describe the effects of soil degradation on crop production		- brainstorming causes of soil degradation - discussing causes of soil degradation - developing cause-effect problem trees on soil degradation - summarizing causes of soil degradation - brainstorm the effects of soil degradation on crop production - discussing the effects of soil degradation on crop production - loss of top soil - reduction of arable land - spread of waterborne and soil borne diseases - spread of weed seeds - silting - flooding - pollution of water sources - developing cause-effect problem trees on the effects of soil degradation on crop production - summarizing the effects of soil degradation on crop production	- brainstorming - discussion - explanation - question and answer - oral questions - problem-trees	- students' experiences - pictures - libraries - charts

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	4 explain the relationship between rapid population growth and soil degradation 5 discuss how to control soil degradation		- researching on the effects of rapid population growth on soil degradation - reporting findings - discussing the relationship between rapid population growth and soil degradation - brainstorming ways of controlling soil degradation - discussing ways of controlling soil degradation - demonstrating ways of controlling soil degradation - practising controlling soil degradation in the local environment	- brainstorming - discussion - research - presentation - explanation - question and answer - oral questions - brainstorming - discussion - demonstration - practical exercises - explanation - question and answer - oral question - written assessment	- students' experiences - libraries - charts - pictures - ICT - local environment - local environment - equipment/materials for controlling soil degradation - students' experiences - resource persons
Assessment	Success	Theme/	Suggested teaching, learning and	Suggested	Suggested

standard	criteria	topic	assessment activities	teaching, learning and assessment methods	teaching and learning resources
We will know this when the students are able to: demonstrate an understanding the ways of dealing with climate change in agriculture	The student must be able to: 1 list ways of dealing with climate change in agriculture	Agriculture and climate change	- researching on ways of dealing with climate change in agriculture - conservation agriculture - water harvesting - agro forestry - re-afforestation - integrated forest, crops, livestock and fish systems - reporting findings - summarizing ways of dealing with change in agriculture	- brainstorming - research - presentation - explanation - oral questions	- local environment - libraries - ICT - chart paper - students' experiences
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources

	2 describe ways dealing with climate change in agriculture		- brainstorming the meaning of the following terms: - conservation agriculture - water harvesting - agro forestry - re-afforestation - integrated forest, farm and fish systems - discussing the meaning of the following terms: - conservation agriculture - agro forestry - water harvesting - re-afforestation - integrated forest, farm and fish systems - outlining the activities done in each way of dealing with climate change	- brainstorming - discussion - explanation - question and answer - research	- local environment - students' experiences - textbooks - ICT
	3 explain how each measure can deal with climate change		- researching on how each measure deals with climate change - discussing how each measure deals with climate change - reporting findings - summarizing how each measure deals with climate change in agriculture	- brainstorming - discussion - explanation - question and answer - research - presentation - oral questions	- local environment - libraries - ICT - chart paper - students' experiences
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this	The students				

when the students are able to:	must be able to :				
demonstrate an understanding of land drainage	1 state the meaning of the term <i>land drainage</i> 2 explain the importance of land drainage 3 describe methods of land drainage	Land drainage	- brainstorming the meaning of the term <i>land drainage</i> - discussing the meaning of the term <i>land drainage</i> - brainstorming the importance of land drainage - discussing the importance of land drainage. - reporting the findings - suggesting methods of land drainage - listing methods of land drainage - surface - subsurface - biodrainage - describing each method of land drainage	- brainstorming - discussion - explanation - oral questions - question and answer - brainstorming - discussion - presentation - explanation - oral questions - question and answer	- students' experiences - libraries - ICT - charts - students' experiences - resource persons - charts - written questions - written reports - pictures

Core element: Agricultural research and technology

Outcome: The students will be able to apply research skills and apply improved agricultural technology to increase farm productivity.

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and	Suggested teaching and
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				assessment methods	learning resources
We will know this when the students are able to: demonstrate an understanding of farm machinery	The students must be able to: - define the meaning of the term <i>farm mechanisation</i> - list types of farm machinery	Farm mechanisation	- brainstorming the meaning of the term <i>farm mechanisation</i> - visiting a mechanized farm to identify types of farm machinery - reporting findings - listing types of farm machinery	- brainstorming - discussion - excursion - written tests - oral assessment - discussion - excursion - written tests - oral assessment - explanation	- students' experiences - pictures - charts - students' experiences - pictures - charts - resource persons - mechanised farm
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	explain the factors to consider when mechanizing a		- brainstorming factors to consider when mechanizing a farm - listing factors to consider when mechanizing a farm	- research - brainstorming - discussion - oral tests	- students' experiences - pictures - charts

	farm		discussing the factors to consider when mechanizing a farm	- written assessment - explanation - question and answer	- resource persons - ICT - libraries
	4 state the advantages of farm mechanisation		- visiting a mechanized farm to observe farm machinery - researching on the advantages of farm mechanization - reporting findings - identifying the advantages of farm mechanization - discussing the advantages of farm mechanization - summarizing the advantages of farm mechanization	- research - brainstorming - discussion - excursion - oral tests - written assessment - explanation	- students' experiences - pictures - charts - resource persons - mechanised farms - ICT - libraries
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	5 state the limitations of farm mechanisation		- visiting a mechanized farm to observe farm machinery - researching on the limitations of farm mechanisation - discussing the limitations of farm mechanisation - reporting findings	- research - brainstorming - discussion - excursion - written exercises - explanation - oral assessment	- students' experiences - checklists - ICT - libraries - pictures - mechanised

	6 describe the maintenance of various farm machinery 7 discuss safety measures when using farm machinery		• summarizing the limitations of farm mechanisation • brainstorming ways of maintaining farm machinery • listing ways of maintaining farm machinery • discussing ways of maintaining farm machinery • brainstorming safety measures when handling farm machinery • discussing safety measures when handling farm machinery • demonstrating safety measures when handling farm machinery • practising safety measures when handling farm machinery	• written tests • research • brainstorming • discussion • question and answer • explanation • oral/written assessment	farms • students' experiences • farm • charts • pictures • diagrams • ICT • libraries • resource persons
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: • demonstrate an understanding of efficient utilization of farm power	The students must be able to: 1 describe sources farm power	Farm power	• brainstorming sources of farm power in Malawi eg – human – animal – wind – mechanical	• brainstorming • discussion • excursion • explanation • research	• students' experiences • checklists • pictures • ICT • libraries

			– water – solar – bio-gas • discussing how different sources of power are used on a farm • visiting nearby farms to observe sources of power in use		• resource persons
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	2 describe the advantages of different sources of farm power		• recalling advantages of different sources of farm power • discussing advantages of different sources of farm power	• discussion • explanation • question and answer • debate • oral assessment • written tests	• students experiences • diagrams • pictures • charts
	3 describe the limitations of different		• brainstorming the limitations of different sources of farm power • discussing the limitations of	• discussion • explanation • question and answer	• students' experiences • diagrams

	sources of farm power		different sources of farm power	• debate • oral/written assessment	• pictures • charts
	4 state ways of improving output from the different sources of farm power		• suggest ways of improving output from the different sources of farm power • discussing ways of improving output from the different sources of farm power	• brainstorming • debate • discussion • question and answer • explanation • research • oral/written assessment	• students' experiences • diagrams • pictures • charts • libraries • ICT • test items

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: • demonstrate an understanding of gender in agricultural technology	The students must be able to: 1 identify gender biases in agricultural technology	Gender and agricultural technology	• brainstorming gender bias in agricultural technology • discussing gender bias in agricultural technology - males driving tractors and oxen - women processing farm produce • summarising gender bias in agricultural technology	• brainstorming • discussion • question and answer • explanation • research • oral/written assessment • case studies	• students' experiences • diagrams • pictures • charts • libraries • ICT • test items

	2 explain causes of gender biases in agricultural technology		- brainstorming causes of gender bias in agricultural technology - discussing causes of gender bias in agricultural technology - consolidating causes of gender bias in agricultural technology	- brainstorming - discussion - question and answer - explanation - research - oral/written assessment - case studies	- students' experiences - diagrams - pictures - charts - libraries - ICT - test items
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 examine the effects of gender bias in agricultural technology		- identifying the effects of gender bias in agricultural technology - debating the effects of gender bias in agricultural technology - low agricultural production - developing future's wheels on the effects of gender bias in agricultural technology	- brainstorming - discussion - question and answer - explanation - research - oral/written assessment - case studies - future's wheels	- students' experiences - diagrams - pictures - charts - libraries - ICT - test items
	4 describe ways of dealing with gender bias in		- brainstorming ways of dealing with gender bias in agricultural technology - discussing ways of dealing with	- brainstorming - discussion - question and answer	- students' experiences - charts - libraries

	agricultural technology		gender bias in agricultural technology consolidating ways of dealing with gender bias in agricultural technology	- explanation - research - oral/written assessment - case studies	- ICT - test items
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of how improved farming technology impacts on food security	The students must be able to: - explain the meaning of the term <i>improved farming technology</i> - explain how improved farming technology	Improved farming technology	- observing examples of farming technology - classifying technology into improved and unimproved technology - discussing the meaning of the term <i>improved technology</i> - summarising the meaning of the term <i>improved technology</i> - researching on how improved technology increases food supply for the growing population - reporting findings - discussing how improved technology	- excursions - discussion - case studies - explanation - oral/written assessment - research - discussion - case studies - explanation - presentation	- local environment - charts - pictures - students' experiences - rubrics - local environment - charts - pictures - students'

	increases food supply		increases food supply • summarising how improved farming technology increases food supply	• oral/written assessment	experiences • libraries • ICT
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 explain how improved farming technology affects food security		• brainstorming how improved farming technology affects food security • discussing how improved farming technology affects food security • summarising how improved farming technology affects food security	• research • discussion • explanation • presentation • oral/written assessment	• local environment • charts • pictures • students' experiences • libraries • ICT

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Core element: Agricultural economics and farm business management

Outcome: The students will be able to appreciate how concepts and principles of agricultural economics and farm business management increase profitability of agricultural enterprises.

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of agricultural marketing and trading	The students must be able to: - distinguish between marketing and trading - list marketing channels and agencies	Agricultural marketing and trading	- brainstorming the meaning of the term <i>trading</i> - discussing the meaning of the term <i>trading</i> - discussing the differences between marketing and trading - brainstorming the meaning of the terms - marketing channels - marketing agencies - discussing the meaning of the terms <i>marketing channels</i> and <i>marketing agencies</i> - listing marketing channels and agencies	- brainstorming - discussion - question and answer - role-playing - brainstorming - discussion - oral assessment - written tests	- students' experiences - pictures - samples of farm produce - students' experiences - charts

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 explain the roles of channels and agencies in marketing of agricultural commodities		- researching on the roles of channels and agencies in the marketing of agricultural commodities - reporting findings - summarizing the roles of channels and agencies in the marketing of agricultural commodities	- research - presentation - explanation - role play	- markets - resource persons - learners experiences - charts - libraries - ICT

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	4 discuss marketing costs and marketing margins 5 describe the effects of population distribution on marketing		- brainstorming the meanings of the terms: - marketing costs - marketing margin - discussing the meanings of the terms: - marketing costs - marketing margin - collecting data on prices of commodities from farms to consumers - calculating marketing costs and marketing margins - discussing the relationship between marketing costs and marketing margins - brainstorming the effects of population distribution on marketing - discussing the effects of population distribution on marketing	- brainstorming - discussion - oral questions - practical exercises - excursions - discussion - teacher observation - brainstorming - discussion - explanation - written exercises - research	- students' experiences - local environment - charts - markets - rubrics - resource persons - students' experiences - libraries - ICT
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources

We will know this when the students are able to: demonstrate an understanding of trading of agricultural commodities	The students must be able to: - explain the importance of trading of agricultural commodities - outline ways of improving trading of agricultural commodities	Trading of agricultural commodities	- brainstorming the importance of trading of agricultural commodities at: - community level - national level - international level - discussing the importance of trading of agricultural commodities at: - community level - national level - international level - reporting findings - brainstorming ways of improving trading of agricultural commodities at: - community level - national level - international level - discussing ways of improving trading of agricultural commodities - summarizing ways of improving trading of agricultural commodities	- brainstorming - discussion - presentation - question and answer - research - brainstorming - discussion - question and answer - oral tests	- markets - resource persons - students' experiences - charts - libraries - ICT - students' experiences - libraries - resource persons
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an	The students must be able to: discuss price	Price	brainstorming definitions of	brainstorming	students'

understanding of price elasticity of demand and supply of agricultural commodities	elasticity of demand and price elasticity of supply	elasticity of demand and supply	– price elasticity of demand – price elasticity of supply • discussing definitions of – price elasticity of demand – price elasticity of supply • consolidating the definitions	• discussion • oral questions	• experiences • elastic and non elastic materials
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	2 calculate price elasticity of demand and supply of agricultural		• collecting data on quantities being bought and sold at specific prices of agricultural commodities on a market • calculating price elasticity of demand and supply	• practical exercises • excursion • discussion • teachers'	• local environment • graph paper • charts • students'

	commodities		- plotting graphs showing different degrees of price elasticity of demand and supply - interpret degrees of price elasticity of demand and supply - elastic - unity - inelastic	- observation of the students participation - presentation	- experiences - markets - rubrics
3	explain the implications of price elasticity of demand and supply on agricultural commodities		- researching on the implications of price elasticity of demand and supply on agricultural commodities - reporting the findings - discussing the implications of price elasticity of demand and supply on agricultural commodities - summarizing the implications of price elasticity of demand and supply on agricultural commodities	- research - explanation - discussion - teachers' observation of the students participation - presentation	- local environment - graph paper - charts - students' experiences - markets - rubrics

Core element: Crop production

Outcome: The students will be able to use scientific knowledge and skills of growing selected crops using recommended husbandry practices.

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to:	The students must be able to:				

demonstrate an understanding of crop improvement	1 describe the meaning of crop improvement	Crop improvement	- brainstorming the meaning of <i>crop improvement</i> - reporting findings - discussing the meaning of crop improvement	- brainstorming - discussion - explanation - question and answer - oral assessment - written assessment	- students' experiences - libraries - ICT - Charts
	2 describe the aims of crop improvement		- researching on the aims of crop improvement - reporting findings - discussing the aims of crop improvement	- research - discussion - explanation - presentation - question and answer - oral tests - written assessment - quizzes	- students' experiences - libraries - ICT - resource persons - charts
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 list activities in crop improvement		- visiting an organization involved in crop improvement - observing activities in crop improvement - recording activities in crop improvement - reporting the findings - summarising activities in crop improvement	- excursion - discussion - explanation - presentation - question and answer - oral assessment - written tests	- students' experiences - libraries - ICT - resource persons - chart - pictures - crop

	4 describe the methods of crop improvement		- researching on the methods of crop improvement - reporting research findings - discussing methods of crop improvement - introduction - selection - hybridization	- research - presentation - discussion - explanation - question and answer	improvement institutions - students' experiences - libraries - ICT - resource persons - charts - pictures - diagrams
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of crop processing	The students must be able to: 1 state the importance of crop processing 2 describe the	Crop processing	- brainstorming the importance of crop processing - discussing the importance of crop processing - improve shelf-life - value addition - improve taste - brainstorming the processing of	- brainstorming - discussion - explanation - research - question and answer - oral assessment - written tests - brainstorming	- students' experiences - libraries - ICT - charts - pictures - samples of crop products - students'

	processing of various crops		various crops - researching on the processing of various crops - maize - groundnuts - mushroom - mangoes - vegetables - reporting findings - discussing the processing of the crops	- discussion - explanation - presentation - research - question and answer - oral assessment - written assessment	experiences - libraries - ICT - resource persons - charts - pictures - samples of crop produce and their products
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 discuss how to process a selected crop		- researching on the processing of the selected crop - reporting findings - discussing the processing of selected crops - processing selected crops	- research - practical - discussion - explanation - question and answer - oral assessment - written tests - teachers' observation of the students ability to process the selected crop	- students' experiences - libraries - ICT - resource persons - charts - pictures - samples of crop produce - rubrics - checklists

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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of efficient pasture production and utilization	The students must be able to: - state the meaning of the term <i>pasture</i> - explain the importance of pasture	Pasture	- brainstorming the meaning of the term <i>pasture</i> - discussing the meaning of the term <i>pasture</i> - summarizing the meaning of the term <i>pasture</i> - researching on the importance of pasture - reporting findings - discussing the importance of pasture	- brainstorming - discussion - explanation - question and answer - oral/written assessment - research - presentation - discussion - explanation - question and	- students' experiences - libraries - ICT - charts - pictures - local environment - students' experiences - libraries - ICT - resource

			- animal feed - controls erosion - improves soil structure - adds soil nutrients - reduces pests and diseases	answer • oral/written assessment	persons • charts/pictures • samples of pasture • local environment
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 state the types of pasture		• brainstorming types of pastures • visiting a pastureland to observe different types of pastures • reporting findings • listing types of pasture - indigenous - exotic	• excursion • practical work • presentation • discussion • explanation • question and answer • oral/written assessment	• students' experiences • libraries • ICT • resource persons • charts • pictures • pasture samples
	4 describe methods of pasture establishment		• researching on methods of pasture establishment • reporting on the methods of pasture establishment • discussing methods of pasture establishment - broadcasting - under sowing - over sowing - vegetative propagation	• research • presentation • discussion • explanation • question and answer • oral/written assessment	• students' experiences • libraries • ICT • resource persons • charts • pictures • pasture samples • local

			- drilling • summarizing methods of pasture establishment		environment
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	5 explain the advantages of each method of pasture establishment		• brainstorming the advantages of different methods of pasture establishment • discussing the advantages of different methods of pasture establishment	• brainstorming • discussion • explanation • question and answer • oral/written assessment	• students' experiences • libraries • ICT • resource persons • charts • pictures • pasture planting materials
	6 explain the disadvantages of each method of pasture establishment		• brainstorming the disadvantages of different methods of pasture establishment • discussing the disadvantages of different methods of pasture establishment	• brainstorming • discussion • explanation • question and answer • oral/written assessment	• students' experiences • libraries • ICT • resource persons • charts • pictures • pasture planting

					materials
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	7 state the meaning of the term <i>seed rate</i> 8 describe factors affecting pasture seed rate		- brainstorming the meaning of the term <i>seed rate</i> - discussing the meaning of the term <i>seed rate</i> - summarizing the meaning of the term <i>seed rate</i> - calculate seed rate for different pastures - predicting factors affecting pasture seed rate - discussing the factors affecting pasture seed rate - seed size - soil tilth - growth habit of pasture - method of sowing - purity percentage - germination percentage - summarizing factors affecting pasture seed rate	- brainstorming - discussion - explanation - question and answer - brainstorming - question and answer - oral/written assessment discussion - explanation	- students' experiences - ICT - charts - pictures - students' experiences - libraries - students' experiences - libraries - ICT - resource persons - charts - pictures - pasture seeds - local environment

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	9 describe how to treat pasture seeds 10 explain the importance of each a method of pasture treatment		- researching on the methods of pasture seed treatment - reporting findings - discussing methods of pasture seed treatment - hulling - scarification - inoculation - pelleting - selecting methods for treating pasture seeds - treating pasture seeds - brainstorming the importance of each method of pasture treatment - discussing the importance of each method of pasture treatment - summarizing the importance of each method of pasture treatment	- research - presentation - practical work - question and answer - discussion - explanation - written/oral assessment - teachers' observation of students' ability to treat pasture seeds - brainstorming - discussion - question and answer - written tests	- students' experiences - libraries - ICT - resource persons - charts - pictures - pasture seeds - local environment - materials for seed treatment - students' experiences - libraries - ICT - charts
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources

	11 discuss proper time for pasture establishment 12 explain activities involved in pasture management		• brainstorming proper time for pasture establishment • discussing the proper time for pasture establishment • preparing the land • planting pasture using an appropriate method • researching on activities involved in pasture management • reporting findings • discussing the activities involved in pasture management - applying the correct type and amount of fertilizer - weeding the pasture land - controlling pests and diseases - grazing animals on the pasture land - controlling burning of the pasture in the dry season • summarising the activities involved pasture management	• brainstorming • presentation • practical work • question and answer • oral discussion • explanation • written questions • research • presentation • question and answer • oral/written assessment • discussion • explanation	• students' experiences • libraries • ICT • resource persons • charts • pictures • local environment • planting materials • students' experiences • libraries • ICT • resource persons • charts • pictures • samples of fertilizers and pesticides
Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources

	13 describe grazing systems in pasture management		- visiting the pasture land to observe grazing systems used - reporting findings - discussing grazing systems - zero grazing - rotational grazing - strip grazing - continuous grazing - deferred grazing - discussing the advantages and disadvantages of each grazing system	- excursions - presentation - question and answer - oral/written assessment - discussion - explanation	- students' experiences - libraries - ICT - resource persons - charts - pictures - local environment
	14 explain methods of pasture conservation		- brainstorming methods of pasture conservation - discussing methods of pasture conservation - hay making - silage making - foggage - summarising methods of pasture conservation	- brainstorming - question and answer - oral/written assessment - discussion - explanation	- students' experiences - libraries - ICT - resource persons - charts - pictures

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	15 practising pasture		selecting methods of pasture conservation	- practical work - demonstration	students' experiences

	conservation		- demonstrating how to conserve pasture using a selected method - conserving pasture using a selected method	- question and answer - oral/written assessment - discussion - explanation	- libraries - ICT - resource persons - charts - pictures - local environment - materials for pasture conservation
16	describe the factors that affect the quality of conserved pasture		- researching on factors that affect the quality of hay and silage - reporting findings - discussing t factors that affect the quality of hay and silage - leafiness of pasture - age at cutting of pasture - moisture content - presence of foreign materials - method of storage - weather condition - pasture species used	- research - presentation - question and answer - oral/written assessment - discussion - explanation	- students' experiences - libraries - ICT - resource persons - charts - pictures - local environment - samples of hay/ silage

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding	The students must be ale to: 1 state the	Mango	researching on the importance	research	students'

of mango production	importance of fruits	production	of fruits - reporting the findings - debating the importance of fruits - discussing the importance of fruits - nutritional - economic - socio - consolidating the importance of fruits	- debate - discussion - explanation - question and answer - oral tests - written tests	experiences - libraries - ICT - charts - pictures - posters
	2 identify different mango varieties		- brainstorming characteristics of different mango varieties - tasting mango fruit - discussing characteristics of different mango	- brainstorming - discussion - explanation - question and answer - oral questions	- students' experiences - charts - samples of mangoes - pictures

Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 select a suitable site for mango production		- brainstorming factors to consider when selecting a site for mango production - identifying a suitable sites for mango production - reporting findings - discussing factors to consider when	- excursion - brainstorming - discussion - explanation - question and answer - oral tests	- students' experiences - libraries - ICT - local environment - charts

	4 prepare a site for planting mango seedlings		- selecting a site for mango production - clearing land - making planting holes - demonstrating applying manure - applying manure	- written tests - discussion - explanation - question and answer - oral tests - written assessment	- students experiences - library - ICT - resource persons - charts - farm tools
	5 demonstrate the planting mango seedlings		- demonstrating planting mango seedlings - planting mango seedlings - demonstrating mulching around mango seedlings - mulching around mango seedlings	- demonstration - explanation - question and answer - oral tests - written assessment - teacher 's observation	- mulching materials - mango orchards - mango seedlings - farm tools - charts - resource persons - students' experiences
Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	6 discuss how to weed the mango orchard		- reviewing weeding methods - selecting appropriate weeding method for mango orchard - demonstrating weeding process - weeding a mango orchard	- practical work - demonstration - discussion - explanation - question and answer - oral tests - written assessment - teacher observation	- students' experiences - orchards - farm tools

	7 identify pests of mangoes		- researching on different pests of mangoes - reporting findings - discussing pests of mangoes - fruit flies - mango scales - mango stone weevil - visiting a nearby mango orchard to observe pests of mangoes - identifying pests of mangoes	of the students participating on weeding - research - discussion - explanation - excursion - question and answer - oral tests - written tests - short essays - teacher observation	- students' experiences - libraries - ICT - charts - orchards - resource persons - pictures
Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	8 discuss how to control pests of mangoes		- brainstorming different methods of controlling mango pests - discussing different methods of controlling mango pests - identifying a particular pest to control using an appropriate method - controlling selected pests	- discussion - explanation - question and answer - oral assessment - written tests - short quizzes - teachers' observation on students participation in	- students' experiences - charts - pesticides
	9 identify		researching on different diseases of		

	diseases of mangoes		mangoes - reporting findings - discussing diseases of mangoes - anthracnose - powdery mildew - visiting nearby mango orchards to observe diseases of mangoes - identifying diseases of mangoes - brainstorming different ways of controlling diseases of mangoes - discussing different ways of controlling diseases of mangoes - identifying a particular disease to control using an appropriate method controlling a selected disease	controlling mango pests - research - excursion - discussion - explanation - oral tests - written tests - presentation - teacher observation on students participation in identifying mango diseases	- libraries - resource persons - charts - orchards - pictures - samples of diseased parts - students experiences - libraries - ICT
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Core element: Livestock production

Outcome: The students will be able to use scientific knowledge and skills of rearing farm animals profitably using appropriate husbandry practices.

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding	The students must be able to: list breeds of cattle for dairy	Breeds of cattle	- researching on breeds of cattle - visiting cattle farms to observe	- research - discussion	nearby livestock units

of dairy and beef production	and beef production		- different breeds of cattle - reporting findings - classifying breeds of cattle into dairy and beef - listing breeds of cattle for beef and dairy production	- presentation - oral assessment - written tests - matching tests	- students' experiences - pictures - charts - library search - web search
	2 describe characteristics of cattle breeds for dairy and beef production		- brainstorming the characteristics of breeds of cattle for beef and dairy production - discussing the characteristics of breeds for beef and dairy production - reporting the findings - consolidating the characteristics of cattle for beef and dairy production	- brainstorming - discussion - presentation - oral assessment - written tests - matching test	- nearby livestock units - students' experiences - pictures/charts - library search

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	3 describe management practices for beef and dairy production		- brainstorming management practices for beef and dairy production - discussing management practices for beef and dairy production	- brainstorming - discussion - presentation - oral assessment - written test - research	- students' experiences - pictures - charts - libraries - ICT - resource persons
	4 identify a suitable house for cattle		- brainstorming on characteristics of a good cattle house - discussing characteristics of a good	- brainstorming - discussion - presentation	- nearby livestock units - students'

			cattle house - visiting a nearby village to observe cattle houses - reporting findings - choosing an appropriate cattle house	- practical - oral assessment - written tests - teachers' observation of students' ability to identify appropriate house - excursion	experiences - pictures - charts - libraries - ICT - resource persons
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	5 discuss how to construct an appropriate house for cattle 6 identify suitable feeds for cattle		- listing materials required for constructing a cattle house - collecting appropriate materials for construction of a cattle house - discussing the procedure for constructing a cattle house - demonstrating how to construct a cattle house - constructing cattle house - brainstorming suitable feeds for cattle - discussing suitable feeds for cattle	- discussion - practical work - oral/written assessment - teachers' observation of the students ability to participate - demonstration - brainstorming - discussion - presentation - practical	- nearby livestock units - students' experiences - pictures - charts - libraries - ICT - nearby livestock units - students' experiences

	7 identify diseases of cattle		• listing suitable feeds for cattle • researching on diseases of cattle • listing diseases of cattle • describing signs of diseases of cattle • role-playing signs of diseases of cattle • discussing signs of diseases of cattle	• oral/written assessment • teachers' observation of students' ability to identify suitable feeds	• pictures • charts • libraries • ICT • resource persons • feed samples
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	8 explain how to control diseases of cattle		• brainstorming control measures of diseases of cattle • discussing control measures of diseases of cattle • reporting findings • explaining control measures of diseases of cattle • observing a resource person controlling diseases of cattle	• brainstorming • discussion • Presentation • oral/written assessment • matching tests • teachers' observation of students' ability to identify diseases	• students' experiences • pictures • charts • libraries • ICT • resource persons
	9 identify parasites of cattle		• researching on parasites of cattle • listing parasites of cattle • describing signs of parasites of cattle • role-playing on the signs of parasite attack in cattle • recording signs of parasites of cattle	• brainstorming • discussion • presentation • practical • oral/written assessment	• nearby livestock units • students' experiences • pictures • charts

	10 explain how to control parasites of cattle		- brainstorming control measures of parasites of cattle - discussing control measures of parasites of cattle - reporting findings - observing a resource person controlling parasites of cattle - summarising control measures of parasites of cattle	demonstration	- libraries - ICT - resource persons - drugs
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching , learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of reproductive systems of poultry and cattle.	The students must be able to: 1 draw reproductive systems of cattle and poultry 2 explain the functions of different parts of the reproductiv	Reproductive systems of poultry and cattle	- researching on reproductive systems of cattle and poultry - drawing reproductive systems of cattle and poultry - brainstorming different parts of the reproductive systems of cattle and poultry - labelling parts of the reproductive systems of cattle and poultry - researching on the functions of different parts of the reproductive systems of cattle and poultry - reporting findings - discussing the functions of	- brainstorming - discussion - presentation - practical work - oral questions - written questions - teachers' observation - research - brainstorming - discussion - presentation - oral/written assessment	- nearby livestock units - students' experiences - pictures/charts - libraries/ICT - abattoirs - abattoirs - students' experiences - pictures/chart s

	e systems of cattle and poultry		different parts of the reproductive systems of cattle and poultry consolidating the functions of different parts of the reproductive systems of cattle and poultry	- written questions - research	- library/ICT
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of the reproductive cycles of cattle, sheep, goats and pigs	The students must be able to: - state the age at puberty for cattle, sheep, goats and pigs - describe the oestrous cycle for a cow - state the signs of heat in livestock	Reproductive cycles of cattle, sheep, goats, pigs and rabbits	- researching on the age of puberty for cattle, sheep, goats, pigs and rabbits - reporting findings - summarizing the age of puberty for cattle, sheep, goats, pigs and rabbits - researching on the oestrus cycle for a cow - discussing the oestrus cycle of a cow - reporting the findings - researching on the meaning of the term <i>heat</i> in livestock - discussing the meaning of the term	- brainstorming - discussion - presentation - oral/written assessment - matching tests - research - brainstorming - discussion - presentation - oral/written assessment - research - brainstorming - discussion - presentation	- students' experiences - library/ICT - livestock production records - students' experiences - library/ICT - livestock production records - students' experiences - library/ICT

			<i>heat</i> in livestock - researching on signs of heat in livestock - discussing signs of heat in livestock - reporting on signs of heat in livestock	- oral/written assessment - research	- livestock production records - animals on heat
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Assessment standard	Success criteria	Theme/ topic	Suggested teaching and learning activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
	4 state the gestation period for cow, ewe, sow, and nanny 5 describe processes of reproduction in cattle		- researching on the gestation period for cow, ewe, sow, and nanny - visiting nearby farms to find out the gestation period for cow, ewe, sow, and nanny - reporting findings - recording the gestation period for cow, ewe, sow, and nanny - researching on the processes of reproduction in cattle - reporting findings - discussing the process of reproduction in cattle - mating - fertilization - embryo development - foetus development - parturition	- brainstorming - discussion - presentation - oral/written assessment - matching tests - research - discussion - presentation - oral/written assessment - matching tests - research	- students' experiences - libraries/ICT - livestock records - students' experiences - library/ICT - local environment - libraries - resource persons

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methods	Suggested teaching and learning resources
We will know this when the students are able to: demonstrate an understanding of livestock improvement	The students must be able to: 1 describe livestock improvement 2 state the aims of livestock improvement	Livestock improvement	- researching on the meaning of the term <i>livestock improvement</i> - reporting findings - discussing the meaning of the term <i>livestock improvement</i> - summarizing the meaning of the term livestock improvement - brainstorming the aims of livestock improvement - discussing the aims of livestock improvement - stating the aims of livestock improvement	- discussion - presentation - oral/written assessment - research - brainstorming - discussion - oral/written assessment	- students' experiences - libraries - ICT - resource persons - students' experiences - libraries - ICT

Assessment standard	Success criteria	Theme/ topic	Suggested teaching, learning and assessment activities	Suggested teaching, learning and assessment methodologies	Suggested teaching and learning resources
	3 describe the methods of livestock improvement 4 identify characteristics of livestock to be selected for breeding 5 describe livestock breeding systems		- researching on methods of livestock improvement - reporting findings - discussing the methods of livestock improvement - breeding - introduction of exotic breeds - summarising methods of livestock improvement - brainstorming characteristics of livestock to be selected for breeding - discussing characteristics of livestock to be selected for breeding - summarizing characteristics of livestock to be selected for breeding - researching on livestock breeding systems - discussing livestock breeding systems - cross breeding - inbreeding - out breeding - reporting findings - summarizing livestock breeding systems	- discussion - presentation - oral/written assessment - research - brainstorming - discussion - oral/written assessment - matching tests - brainstorming - discussion - presentation - oral questions - written question - research - case studies	- students' experiences - libraries - ICT - resource persons - students' experiences - libraries - ICT - resource persons - charts/pictures - students' experiences - libraries - ICT - local environment - resource persons - charts/pictures

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